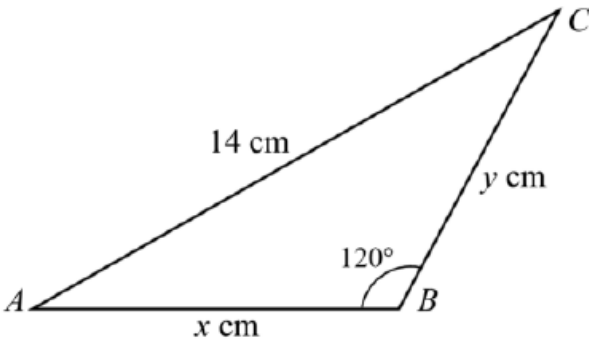


1	ACSB/2025/I/09
	(a) Explain why the line $y = 3x + k$ will always intersect the quadratic curve $y = kx^2 + 2x + 1$ for all values of k, where k is a constant. [4] (b) If the line $y = 3x + k$ is the tangent to the quadratic curve $y = kx^2 + 2x + 1$, find the value of k. [2]
2	ACSB/2025/I/11
	A basketball is thrown vertically upwards. Its height, h m, above the ground at time x seconds after being thrown is given by the formula $h = 1.68 + 7x - 7x^2$. (a) State the height above the ground from which the ball is thrown. [1] (b) Express h in the form $a + b(x + c)^2$ where a, b and c are constants to be determined. [3] (c) Hence state the maximum height attained by the ball and the time at which this occurs. [2] (d) The ball hits the ground. Explain why the time taken for the ball to hit the ground is not twice the time found in part (c). [1]
3	BASS/2025/I/01
	The equation of a curve is $y = -3x^2 + 6x + 5$. (a) By expressing $-3x^2 + 6x + 5$ in the form of $a(x + b)^2 + c$ where a, b, and c are constants, find the coordinates of the stationary point on the curve. [2] (b) Hence explain why $-3x^2 + 6x + 5$ is always less than 10. [2]
4	BASS/2025/I/04
	(a) Find the range of values of k for which $y = 3k(x - 1)$ cuts the curve $y = x^2 + 2kx - 5$ at two distinct points. [5] (b) Determine the conditions of p and q such that the curve $y = px^2 - 2x + 3q$ lies entirely above the x-axis, where p and q are constants. [3]
5	CCHY/2025/II/216/8/2026
	The equation of a curve is $y = px^2 - (p + 2)x + 1$, where p is a constant. (a) Given that $p > 0$, by completing the square, find the minimum value of y. Express your answer in the form $\frac{a - p^2}{bp}$, where a and b are integers. [3] (b) Given instead that $p < 0$, find the coordinates of the maximum point in terms of p. [2]

6	GMS/2025/I/9
	The height, y m, of a ball above the ground after it has been thrown is given by $y = -\frac{1}{16}x^2 + \frac{9}{8}x + 2$ where x m is the horizontal distance travelled. (a) Express the function in the form $y = a(x-h)^2 + k$. [2] (b) Find the maximum height that the ball can reach. [1] (c) A goal post is positioned 22 m away from where the ball was thrown. Justify if the ball can reach the goal post. [2]
7	CCHY/2025/I/4
	In triangle ABC below, $AB = x$ cm, $BC = y$ cm, $AC = 14$ cm and angle $ABC = 120^\circ$.  (a) Using cosine rule, form an equation involving x and y. [1] (b) Given that the perimeter of triangle ABC is 30 cm, find the exact area of triangle ABC. [4]
8	CCHY/2025/I/8
	(a) Find the value of a and of b for which $\{x : x < -3.5 \text{ or } x > 2\}$ is the solution set of $b < x^2 + ax$. [3] (b) Find the range of values of p for which $y = x^2 - px - 3x - p$ is always positive. [3] (c) Explain whether the line $y = -5x - 2$ intersects the curve $y = kx^2 + 3$ where $k < 1$. [3]
9	GMS/2025/II/6
	The equation of a curve is $y = x^2 + (3k-1)x + 2k + 10$. (a) Show that the x-axis is a tangent to the curve when $k = 3$. [2] (b) Find the range of values of k for which y is always positive. [4]

1	(a) $kx^2 + 2x + 1 = 3x + k$ $kx^2 - x + 1 - k = 0$ $D = b^2 - 4ac$ $= 1 - 4k + 4k^2$ $= (2k - 1)^2$ Since $D > 0$ for all values of k, the curve has 2 real and distinct roots. Hence the line will always intersect the curve. (b) line is tangent to the curve $\Rightarrow D = 0$ $b^2 - 4ac = 0$ $(2k - 1)^2 = 0$ $k = \frac{1}{2}$
2	(a) $h = 1.68\text{m}$ $h = -7 \left[x^2 - \frac{x}{2} + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \right] + 1.68$ (b) $= -7 \left(x - \frac{1}{2} \right)^2 + 7.25 + 1.68$ $= 3.43 - 7 \left(x - \frac{1}{2} \right)^2$ (c) Max height = 3.43m Time = 0.5 seconds (d) The ball was not thrown from the ground.
3	1a $-3x^2 + 6x + 5 = -3\left(x^2 - 2x - \frac{5}{3}\right)$ $= -3\left[(x - 1)^2 - 1 - \frac{5}{3}\right]$ $= -3\left[(x - 1)^2 - \frac{8}{3}\right]$ $= -3(x - 1)^2 + 8$ Coordinate: (1, 8) 1b Since max value of $-3x^2 + 6x + 5 = 8$ Therefore $-3x^2 + 6x + 5 \leq 8 < 10$ $-3x^2 + 6x + 5 < 10$

4	4a Sub $y = x^2 + 2kx - 5$ into $y = 3k(x-1)$ $x^2 + 2kx - 5 = 3k(x-1)$ $x^2 - kx + 3k - 5 = 0$ Since the line intersects the curve at two distinct points, Discriminant > 0 $(-k)^2 - 4(1)(3k-5) > 0$ $k^2 - 12k + 20 > 0$ $(k-2)(k-10) > 0$ Range $k < 2$ or $k > 10$ 4b Since curve lies entirely above line, $b^2 - 4ac < 0$ and $p > 0$ $(-2)^2 - 4p(3q) < 0$ $4 - 12pq < 0$ $pq > \frac{1}{3}$ $p > 0$ and $pq > \frac{1}{3}$	
5	(a) $y = px^2 - (p+2)x + 1$ $= p\left(x^2 - \frac{p+2}{p}x\right) + 1$ $= p\left[x^2 - \frac{p+2}{p}x + \left(\frac{p+2}{2p}\right)^2 - \left(\frac{p+2}{2p}\right)^2\right] +$ $= p\left[\left(x - \frac{p+2}{2p}\right)^2 - \left(\frac{p+2}{2p}\right)^2\right] + 1$ $= p\left(x - \frac{p+2}{2p}\right)^2 + 1 - \frac{p^2 + 4p + 4}{4p^2} \times p$ $= p\left(x - \frac{p+2}{2p}\right)^2 + \frac{4p - p^2 - 4p - 4}{4p}$ $= p\left(x - \frac{p+2}{2p}\right)^2 + \frac{-p^2 - 4}{4p}$ Hence, the minimum value of y is $\frac{-p^2 - 4}{4p}$. (b) $\left(\frac{p+2}{2p}, \frac{-p^2 - 4}{4p}\right)$	

6**(a)**

$$y = -\frac{1}{16}x^2 + \frac{9}{8}x + 2$$

$$y = -\frac{1}{16}(x^2 - 18x) + 2$$

$$y = -\frac{1}{16}[(x-9)^2 - (-9)^2] + 2$$

$$y = -\frac{1}{16}(x-9)^2 + \frac{81}{16} + 2$$

$$y = -\frac{1}{16}(x-9)^2 + 7\frac{1}{16}$$

(b)

$$\text{Max Height} = 7\frac{1}{16} \text{ or } 7.0625 \text{ (Do not accept 7.06 since the answer is exact)}$$

(c)

$$0 = -\frac{1}{16}x^2 + \frac{9}{8}x + 2$$

$$0 = -x^2 + 18x + 32$$

$$0 = x^2 - 18x - 32$$

$$x = \frac{18 \pm \sqrt{(-18)^2 - 4(1)(-32)}}{2(1)}$$
$$= \frac{18 \pm \sqrt{452}}{2}$$

$$x = 19.63 \text{ or } x = -1.63$$

Since the ball will land at $x = 19.6\text{m}$, it will miss the goal post which is positioned

at $x = 22\text{ m}$ away.

[Answer +Explanation required]

6**(a)**

$$14^2 = x^2 + y^2 - 2xy \cos 120^\circ$$

$$196 = x^2 + y^2 + xy$$

(b)

$$x + y + 14 = 30$$

$$x = 16 - y \quad \text{-----} [1]$$

$$196 = x^2 + y^2 + xy \quad \text{-----} [2]$$

sub [1] into [2]

$$196 = (16 - y)^2 + y^2 + (16 - y)y$$

$$196 = 256 - 32y + y^2 + y^2 + 16y - y^2$$

$$y^2 - 16y + 60 = 0$$

$$(y - 6)(y - 10) = 0$$

$$y = 6 \quad \text{or} \quad y = 10$$

$$x = 10 \quad \text{or} \quad x = 6$$

$$\text{Area of } \triangle ABC = \frac{1}{2}(10)(6) \sin 120^\circ$$

$$= 30 \left(\frac{\sqrt{3}}{2} \right)$$

$$= 15\sqrt{3} \text{ cm}^2$$

(a)

$$(2x+7)(x-2) > 0$$

$$2x^2 + 3x - 14 > 0$$

$$x^2 + ax - b > 0$$

$$2x^2 + 2ax - 2b > 0$$

$$2a = 3$$

$$a = 1.5$$

$$2b = 14$$

$$b = 7$$

$$x^2 + 1.5x > 7$$

(b)

$$b^2 - 4ac < 0$$

$$(-p-3)^2 - 4(1)(-p) < 0$$

$$p^2 + 6p + 9 + 4p < 0$$

$$p^2 + 10p + 9 < 0$$

$$(p+1)(p+9) < 0$$

$$-9 < p < -1$$

(c)

$$-5x - 2 = kx^2 + 3$$

$$kx^2 + 5x + 5 = 0$$

$$b^2 - 4ac = 25 - 4k(5)$$

$$= 25 - 20k$$

$$k < 1$$

$$-20k > -20$$

$$25 - 20k > 5$$

$$> 0$$

Since $b^2 - 4ac > 0$, for $k < 1$, the line intersects the curve.

8**(a)**When $k = 3$,

$$y = x^2 + 8x + 16$$
$$= (x + 4)^2$$

The curve has a repeated root at $x = -4$ The $x =$ axis is a tangent to the curve.**(b)**

$$x^2 + (3k - 1)x + 2k + 10 > 0$$

$$b^2 - 4ac < 0$$

$$(3k - 1)^2 - 4(1)(2k + 10) < 0$$

$$9k^2 - 6k + 1 - 8k - 40 < 0$$

$$9k^2 - 14k - 39 < 0$$

$$(9k + 13)(k - 3) < 0$$

$$-\frac{13}{9} < k < 3$$